

Gravity as an Emergent Effect of Projected Radiation

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Abstract

This paper proposes that gravity is not a fundamental interaction but an emergent effect arising from the interaction between projected matter and a background radiation field within the SxD dimensional cascade. Drawing on a prior analysis of the matter-radiation classificatory inconsistency in modern physics, and on the mass-energy equivalence $E = mc^2$, it is argued that mass is condensed radiation and radiation is released mass. Within the SxD framework, observable three-dimensional spacetime (S^3D) is a structural projection from a six-dimensional primordial nucleus (S^6D), and matter is a high-density condensation of the projected radiation field. A body of projected mass necessarily perturbs the ambient radiation field; a second body experiences an anisotropic flux, a radiation shadow, producing a net directional force. Under two minimal postulates, that radiative source strength and coupling cross-section are each proportional to mass in the weak-field limit, the Newtonian inverse-square law $F = G_{\text{eff}} Mm/r^2$ emerges from spherical flux geometry. The radiation potential recovers the classical Newtonian potential, and the weak-field metric perturbation reproduces the linearised general-relativistic result. The stress-energy tensor is reinterpreted as the S^3D representation of the projected radiation field, and the metric curvature of general relativity is proposed as an effective geometric encoding of the underlying radiative process. The equivalence principle is derived as a structural consequence rather than an independent postulate. Three conditions for empirical distinguishability and the principal limitations, including the absence of explicit non-linear radiation-field dynamics, are stated. Newton described the phenomenon; Einstein described the geometry; this paper proposes the source.

Keywords: emergent gravity; projection cosmology; dimensional cascade; radiation field; Newtonian limit; general relativity; mass-energy equivalence; ontology of spacetime; SxD model; philosophy of gravity

I. Introduction: The Missing Cause of Gravity

Gravity is the oldest recognised force and the least understood. Newton's Principia (1687) established that every massive body attracts every other with a force proportional to the product of their masses and inversely proportional to the square of their separation [1]. This law describes what gravity does with extraordinary precision but says nothing about why it does it. Newton himself acknowledged this explicitly: "I have not yet been able to discover the cause of these properties of gravity from phenomena, and I frame no hypotheses" [1].

Two centuries later, Einstein's general theory of relativity (1915) reframed the question [2]. Mass does not attract other mass at a distance; rather, mass curves the geometry of spacetime, and objects follow geodesics through that curved manifold. The explanatory gain is substantial: gravitational phenomena are reduced to geometry. Yet the foundational question persists in a new form—why does mass curve spacetime? General relativity provides no answer. The Einstein field equations, $G_{\mu\nu} = 8\pi G T_{\mu\nu}$, assert a correspondence between the energy-momentum content of spacetime and its geometric curvature, but the correspondence is postulated, not derived from a deeper mechanism [3].

The programme of quantum gravity—initiated by Rosenfeld (1930) and Bronstein (1936) and pursued for nearly a century through string theory, loop quantum gravity, and related frameworks—has sought to unify gravitation with quantum mechanics by quantising the gravitational field itself [4,5]. This programme rests on an unstated assumption: that because the electromagnetic field admits quantisation [17,18], the gravitational field must admit it as well. Yet gravity and electromagnetism differ in a fundamental respect. Electromagnetic quanta (photons) have been observed directly and repeatedly. No graviton has ever been detected, and no experiment has ever revealed a discrete quantum of gravitational interaction. The analogy between the two fields, while mathematically suggestive, has no empirical foundation. After ninety years of effort, quantum gravity remains without a single experimentally confirmed prediction.

The scale of investment in this programme deserves emphasis. String theory, the most prominent approach to quantum gravity, has engaged thousands of theoretical physicists over five decades and has generated a landscape of approximately 10^{500} possible vacuum solutions—none of which has yet produced an experimentally confirmed prediction at accessible energies [6]. Loop quantum gravity, the principal alternative, predicts a discrete structure of spacetime at the Planck scale (approximately 10^{-35} metres)—a regime inaccessible to any conceivable experiment. The empirical sterility of these programmes is not a temporary setback; it may be a structural signal that the foundational premise—gravity must be quantised because electromagnetism can be—is incorrect.

More recently, alternative approaches have explored the possibility that gravity is not fundamental but emergent. Jacobson (1995) derived the Einstein field equations from thermodynamic considerations

applied to local Rindler horizons [7]. Verlinde (2011) proposed that gravity arises as an emergent force associated with information on holographic screens [8]. Padmanabhan has independently developed a systematic case that the Einstein equations have the character of emergent thermodynamic relations rather than fundamental field equations [9]. These programmes share the insight that gravity may not require quantisation because it may not be a fundamental field interaction at all. Independently, the holographic principle, arising from the discovery that a black hole's information capacity is determined by its surface area rather than its volume [10,11,12], has established that the information content of a spatial region is determined by its boundary, not its interior. This result, while not invoked as a foundation for the present model, is independently consistent with an ontology in which three-dimensional space is a projection surface rather than a fundamental container. The present paper shares this premise but proposes a distinct mechanism grounded in the SxD dimensional cascade framework [13,14].

The SxD model, formalised in Ripple-Instantiation Cosmogenesis [13], proposes that observable three-dimensional spacetime (S^3D) is not ontologically fundamental but is generated as a structural projection from a six-dimensional primordial nucleus (S^6D) through a cascade of intermediate dimensional layers: $S^6D \rightarrow S^5D \rightarrow S^4D \rightarrow S^3D$. In this framework, matter is not an independent substance occupying space; it is a condensed, high-density region within a pervasive radiation field that constitutes the projection medium itself. The radiation field is the ontological ground; matter is derivative.

If matter is embedded within a radiation field—as a fish is embedded within the sea—then the presence of a massive body necessarily perturbs the radiation environment surrounding it. A second body, immersed in the same field, experiences an anisotropic radiation flux: the first body partially shadows or restructures the radiation arriving from one direction, creating a net directional pressure. In S^3D , this directional pressure is observed as gravitational attraction.

The remainder of this paper develops this proposal in six stages. Section II establishes the ontological framework and the identification of matter as condensed radiation, drawing on both the SxD dimensional cascade and the mass-energy equivalence of special relativity. Section III formalises the relationship between projected mass, radiative source strength, and coupling cross-section. Section IV derives the Newtonian force law from radiative flux geometry and addresses the mechanism of radiation shadowing. Section V establishes the relationship between the radiation-field model and general relativity through the energy-momentum tensor and the effective metric correspondence. Section VI discusses the status of quantum gravity within this framework. Section VII identifies conditions for empirical distinguishability and states the principal limitations.

II. Matter as Condensed Radiation: Ontological and Physical Foundations

II.1 The Matter-Radiation Distinction Reconsidered

The classical ontological opposition between matter and radiation—matter as localised, massive, and substantial; radiation as propagating, massless, and immaterial—has been progressively undermined by the development of quantum field theory. In a prior analysis [15], it was shown that the coexistence of the Standard Model's fermion/boson taxonomy with the inherited nineteenth-century matter/radiation terminology produces a genuine classificatory inconsistency. Beta radiation is an electron flux—the electron being a fermion conventionally classified as a matter particle. Gamma radiation is a photon flux—the photon being a massless gauge boson classified as a force carrier. The single term "radiation" thus encompasses entities that the Standard Model places in ontologically distinct categories. The inconsistency is not a matter of imprecise language; it reflects a failure to revise foundational terminology in light of the conceptual transformation wrought by quantum field theory.

Within the Standard Model itself, the distinction between matter and radiation is not categorical but modal. Both the electron and the photon are described as localised excitations of quantum fields. They differ in the type of field, the quantum numbers carried, and the mode of excitation, but they share the same fundamental ontological status: field excitations. The dynamical evidence confirms this. Pair production (a photon of sufficient energy produces an electron-positron pair) and annihilation (an electron and positron annihilate into photons) demonstrate that the transition between what classical terminology calls "radiation" and what it calls "matter" is empirically routine, governed by conservation laws, and requires no exotic physics. An entity classified as radiation (a photon) becomes an entity classified as matter (an electron) and vice versa.

This paper adopts the conclusion of that analysis: the ontological opposition between matter and radiation is not sustained by modern physics. Matter and radiation are not two fundamentally different kinds of thing. They are two manifestations of the same underlying reality—quantum field excitations—differing in mode, not in kind.

II.2 Mass-Energy Equivalence as a Radiative Identity

The identification of matter with condensed radiation receives independent support from the most famous equation in physics. Einstein's special relativistic mass-energy equivalence, $E = mc^2$, establishes that mass and energy are not merely correlated but are interconvertible manifestations of the same physical quantity [16]. A body of rest mass m possesses an intrinsic energy content $E = mc^2$; conversely, any concentration of energy E possesses an effective mass $m = E/c^2$.

Quantum field theory further establishes that radiation is a form of energy. A photon of frequency ν carries energy $E = h\nu$ [17,18]. Radiation is not merely associated with energy; it is energy in its propagating, delocalised form.

The conjunction of these two established results points toward a conclusion whose dynamical core is already confirmed within the Standard Model itself—in quantum chromodynamics, the overwhelming majority of hadronic mass arises not from intrinsic constituent masses but from the binding energy of massless gluon fields [19]. What has not been drawn from this fact is its full ontological consequence: if mass is energy ($E = mc^2$) and radiation is energy ($E = h\nu$), then the identification of mass as condensed radiation and radiation as released mass becomes a natural ontological inference. This inference is not a deductive consequence of special relativity alone; it requires the additional ontological commitment, supplied by the SxD framework, that the radiation field is the fundamental substrate from which mass condenses:

$$E^2 = p^2c^2 + m^2c^4$$

For a massless particle ($m = 0$), this reduces to $E = pc$ —the standard energy-momentum relation for radiation. For a massive particle at rest ($p = 0$), it reduces to $E = mc^2$. The two cases are not ontologically distinct regimes; they are limiting cases of the same continuous energy-momentum relation. The massive case represents energy in a bound, localised configuration—condensed radiation. The massless case represents energy in a propagating, delocalised configuration—free radiation. The invariant mass term m^2c^4 on the right-hand side of the full relation represents, in this reading, the energy content locked into the bound configuration of the field excitation.

Within the SxD framework, this reinterpretation acquires a structural ground. Mass is not an independent fundamental quantity; it is the S³D effective measure of the degree of radiative condensation at a given projected location. A body of mass M is a region of the projected radiation field where the radiation density is locally elevated into a self-sustaining bound configuration. The greater the mass, the more radiation is condensed at that location.

II.2.1 The Photon-Electron Continuum and the Meaning of Invariant Mass

The full relativistic energy-momentum relation deserves closer examination, because it reveals something that the simplified $E = mc^2$ obscures. Consider an electron at rest: its energy is $E = m_e c^2 = 0.511$ MeV. Now consider a gamma-ray photon with the same energy, 0.511 MeV: its momentum is $p = E/c$, and its rest mass is zero. In the Standard Model, these are classified as fundamentally different entities—the electron is a fermion, a matter particle; the photon is a boson, a force carrier. Yet pair production converts a photon of energy greater than or equal to 1.022 MeV into an electron-positron pair, and the reverse process (annihilation) converts the pair back into photons. The total energy is conserved; the total momentum is conserved; but the classification of the entity changes from "radiation" to "matter" and back.

Within the present framework, this is not a mysterious transmutation between ontological categories. It is a change in the mode of a field excitation: from a propagating, delocalised mode (radiation) to a bound, localised mode (matter), and back. The invariant mass term m^2c^4 in the relation $E^2 = p^2c^2 + m^2c^4$ measures the energy locked into the bound configuration. When the configuration unbinds (annihilation), that energy is released as propagating radiation. When radiation is forced into a bound

configuration (pair production), that energy is locked in as invariant mass. Mass is not a substance; it is a state—the bound state of radiation.

This reading resolves an old puzzle. The question "why does mass have energy?" (answered by $E = mc^2$) is traditionally treated as primitive—mass just does have energy, and the equation tells us how much. But if mass is condensed radiation, the question dissolves: mass has energy because it is energy, in a bound configuration. The equation $E = mc^2$ is not a conversion formula between two different things; it is a statement of identity between two descriptions of the same thing.

II.3 The SxD Dimensional Cascade and the Radiation Sea

The dimensional notation SxD, introduced in [13], denotes six stratified layers of cosmological structure defined not by spatial axes but by their function within the projection cascade. S⁶D is the bounded, spherical primordial nucleus in a state of absolute rest. S⁵D is the atemporal five-dimensional configuration manifold receiving the direct radiation of S⁶D. S⁴D is the quantum-field conversion interface. S³D is the observable material universe. S²D and S¹D are sub-material layers outside the scope of the present paper. S⁰D is the containing volume—the vacuum—within which all layers are embedded.

The generative event is instantaneous dimensional projection:

$$S^6D \rightarrow S^5D \rightarrow S^4D \rightarrow S^3D \rightarrow S^2D \rightarrow S^1D$$

Within S³D, the observable universe is not empty space populated by material objects. It is a pervasive radiation field—the residual projection medium—within which localised high-density condensations appear as what S³D observers identify as matter. The relationship between the radiation field and the material objects embedded within it is not analogous to that between a container and its contents. It is analogous to that between an ocean and the structures within it: the ocean is primary; the structures are derivative patterns within it. A fish does not exist alongside the sea; it is constituted by the sea and immersed within it. In the same way, a material body does not exist alongside the radiation field; it is a condensed configuration of the radiation field, embedded within the uncondensed remainder.

This ontological ordering has direct physical consequences. If a body of mass M is a region of locally elevated radiation density, then its presence necessarily modifies the radiation environment in its vicinity. The ambient radiation field is not a passive background unaffected by the presence of matter; matter, being condensed radiation, interacts with the field from which it is constituted. This interaction is the physical ground from which gravity will be derived in the subsequent sections.

III. Projected Mass and Radiative Coupling

III.1 The Projected Mass Postulate

The ontological framework established in Section II—matter as condensed radiation, embedded within a pervasive radiation field—entails a specific relationship between a body's mass and its effect

on the ambient radiation environment. If mass is the degree of local radiative condensation, then a body of greater mass represents a greater perturbation of the radiation field. This relationship is formalised in two postulates.

Postulate 1 (Radiative Source Strength).

In the weak-field, far-field regime, the effective radiative perturbation produced by a projected body of mass M is proportional to M :

$$L_R(M) = \alpha \times M$$

where L_R is the effective radiative source intensity and α is a coupling constant encoding the strength of the mass-radiation interaction in S^3D . This postulate is not an ad hoc assumption. It is a direct structural consequence of the identification of mass with radiative condensation: if mass measures the degree of condensation, and the perturbation of the ambient field is produced by that condensation, then the perturbation magnitude scales with the condensation measure.

The linearity of this relationship is explicitly asserted as a first-order approximation valid in the weak-field limit. At very high mass densities—in the vicinity of neutron stars, black holes, or cosmologically dense regions—the perturbation produced by the body is no longer small relative to the background radiation field, and the linear scaling $L_R = \alpha \times M$ ceases to be adequate. The non-linear regime is addressed in Section V.

Postulate 2 (Radiative Coupling Cross-Section).

A test body of mass m couples to the ambient radiation field with an effective cross-section proportional to m :

$$\sigma_R(m) = \beta \times m$$

where β is a coupling constant. This postulate follows from the same ontological ground as Postulate 1: if mass is radiative condensation, then a body's susceptibility to radiative flux is determined by the same quantity that defines its condensation—namely, its mass. A more massive body, being a larger condensation of radiation, presents a proportionally larger effective cross-section to the ambient field.

The proportionality $\sigma_R = \beta \times m$ further presupposes that the shadowing produced by a body does not saturate—that is, that each unit of condensed radiation within the body couples to the ambient flux independently. This weak-coupling condition is the cross-sectional counterpart of the weak-field condition stated for Postulate 1: at extreme condensation densities, where shadowing saturates, the linear scaling fails, and the force ceases to track mass exactly. The breakdown of both postulates in the same regime is not a coincidence but a single non-linearity viewed from the source side and the receiver side respectively.

III.2 Why Both Postulates Follow from a Single Ontological Commitment

It is worth emphasising that the proportionality of both source strength and coupling cross-section to mass is not two independent assumptions but two consequences of one: the identification of mass with

radiative condensation. In standard Newtonian gravity, the equality of gravitational and inertial mass is an empirical fact (the equivalence principle) that has no explanation within the theory itself—it is simply observed to hold. In the present framework, the equivalence is a structural entailment. A body's capacity to perturb the radiation field (its "gravitational charge") and its capacity to respond to the radiation field (its "gravitational susceptibility") are both determined by the same physical quantity: the degree of radiative condensation at that location. They are not merely empirically equal; they are the same thing measured in two different ways. This provides a natural ontological basis for the equivalence principle without requiring it as an independent postulate.

IV. Derivation of the Newtonian Force Law from Radiative Flux Geometry

IV.1 Spherical Flux Propagation

Consider a body of projected mass M situated within the ambient radiation field. By Postulate 1, this body produces a radiative perturbation of effective intensity $L_R(M) = \alpha \times M$. In S^3D , this perturbation propagates outward through three-dimensional space. By the geometry of spherical propagation, the radiative flux at distance r from the source is distributed uniformly over the surface area of a sphere of radius r :

$$\Phi_R(r) = L_R(M) / (4 \pi r^2) = \alpha \times M / (4 \pi r^2)$$

This is not a physical assumption but a geometric identity: any isotropic quantity emitted from a point source in three-dimensional Euclidean space dilutes as $1/r^2$ by conservation of flux through expanding spherical surfaces.

IV.2 Force on a Test Body

Now introduce a second projected body of mass m at distance r from M . By Postulate 2, this body couples to the local radiation flux with cross-section $\sigma_R(m) = \beta \times m$. The effective force experienced by m due to the radiative flux perturbation produced by M is the intercepted momentum flux—radiation flux divided by c yields momentum flux, the classical mechanism of radiation pressure [20]:

$$\begin{aligned} F_R &= \sigma_R(m) \times \Phi_R(r) / c \\ &= (\beta \times m / c) \times (\alpha \times M / 4 \pi r^2) \\ &= (\alpha \times \beta / 4 \pi c) \times Mm / r^2 \end{aligned}$$

Defining the effective gravitational constant as:

$$G_{\text{eff}} = \alpha \times \beta / (4 \pi c)$$

we obtain:

$$F_R = G_{\text{eff}} \times Mm / r^2$$

This is Newton's law of universal gravitation. The inverse-square dependence arises not from an intrinsic property of a "gravitational force" but from the geometric dilution of radiative flux over spherical surfaces in three-dimensional space. The proportionality to Mm arises from the two postulates, both of which are structural consequences of the single ontological commitment that mass is condensed radiation.

IV.3 The Radiation Potential

The force law derived above can equivalently be expressed in terms of a scalar radiation potential. Define:

$$U_R(r) = -G_{\text{eff}} \times M / r$$

Then the gravitational acceleration field is:

$$\mathbf{g}_R(r) = -\nabla U_R = -G_{\text{eff}} \times M / r^2 \times \mathbf{\hat{r}}$$

and the force on a test body m is:

$$\mathbf{F}_R = m \times \mathbf{g}_R = -G_{\text{eff}} \times Mm / r^2 \times \mathbf{\hat{r}}$$

This potential is formally identical to the Newtonian gravitational potential. The physical content of the SxD model lies not in the form of the potential—which is geometrically determined—but in the identification of the physical medium (projected radiation) whose flux produces the potential.

IV.4 The Mechanism of Attraction: Radiation Shadowing

The mathematical derivation of Sections IV.1-IV.3 establishes the force law from radiative flux geometry. The present section provides the physical picture underlying that mathematics. A natural objection arises: if both M and m are immersed in a uniform, isotropic background radiation field, why is there a net force at all? An isolated body in a perfectly isotropic field experiences equal radiation pressure from all directions, yielding zero net force. Why does the background radiation not act as a universal drag or resistance?

The answer lies in the ontological relationship between the body and the field. In the SxD framework, a material body is not a foreign object inserted into an external radiation medium. It is a condensed configuration of the radiation field itself. The relationship between the body and the field is not that of an object immersed in a fluid; it is that of a vortex within a flow, or a knot within a rope. A knot does not experience drag from the rope, because the knot is the rope. The background radiation field does not exert a net force on an isolated body, because the body is the field—locally condensed. There is no interface at which "radiation pressure" could act, because there is no ontological boundary between "body" and "field."

This changes when a second body is present. Body M , being a region of elevated radiative condensation, modifies the ambient radiation field in its vicinity. Specifically, it reduces the radiation flux propagating through its location—it casts a radiation shadow. Body m , located at distance r from M , therefore experiences a radiation environment that is no longer isotropic: the flux arriving from the

direction of M is reduced relative to the flux arriving from all other directions. This asymmetry produces a net directional force on m , directed toward M . In S^3D , this is observed as gravitational attraction.

The analogy with the ocean illuminates the mechanism. A single fish in a uniform ocean current experiences no net force in any particular direction—the current flows through it and around it equally. But when a second, much larger body is present nearby, it disrupts the uniformity of the current: the flow arriving at the fish from the direction of the larger body is partially blocked or diverted. The resulting asymmetry pushes the fish toward the larger body—not because the larger body pulls, but because the current behind the fish, unobstructed, pushes harder than the current between them.

This mechanism also explains why the gravitational force is always attractive and never repulsive: a massive body can only reduce the radiation flux in its shadow, never increase it. The asymmetry is therefore always directed toward the perturbing body. This ontological point also distinguishes the present model from the historical Fatio-Le Sage tradition of push-gravity [21]. Le Sage's mechanism posited a flux of ultramundane corpuscles mechanically impacting material bodies, and it fails on two well-known grounds: a body moving through the corpuscular flux experiences a velocity-dependent drag incompatible with the observed stability of planetary orbits, and the absorbed momentum implies an energy deposition that would heat the body catastrophically. Both objections presuppose a mechanical interface between body and flux—a surface at which corpuscles strike and are absorbed. In the present model no such interface exists. The body is not an obstacle within the field but a condensed configuration of the field; the shadowing effect is a restructuring of the field by its own condensation, not an absorption of momentum-carrying projectiles by a foreign object. The drag and heating objections therefore do not transfer: they refute a mechanical corpuscular model, not a field-ontological one. What Le Sage guessed at the level of mechanism, the present model grounds at the level of ontology.

IV.5 The Equivalence Principle as a Structural Consequence

The equivalence principle—the empirical identity of gravitational and inertial mass—is one of the most precisely tested facts in physics (torsion-balance experiments confirm it to parts in 10^{13} and the MICROSCOPE space mission to parts in 10^{15} [22]) and one of the least explained. In Newtonian gravity, it is an unexplained coincidence. In general relativity, it is elevated to a postulate: the equivalence principle is assumed, not derived, and the entire geometric structure of the theory is built upon it.

In the radiation-field model, the equivalence principle is neither a coincidence nor a postulate. It is a structural consequence of the single ontological identification of mass with radiative condensation (Section III.2). A body's gravitational mass—its capacity to produce a radiation-field perturbation—is proportional to its radiative condensation. Its inertial mass—its resistance to acceleration—is also proportional to its radiative condensation, because acceleration of a condensed radiation structure

requires work against the binding energy of that condensation. Both quantities are manifestations of the same underlying physical property. Their equality is not remarkable; their inequality would be.

This constitutes a genuine explanatory gain over both Newtonian gravity (where the equivalence is unexplained) and general relativity (where it is assumed). The radiation-field model provides a structural ontological basis for the equivalence principle from a single commitment, rather than requiring it as an independent foundational postulate.

IV.6 Comparison: Newton, Einstein, and the Radiation-Field Model

Newton described the phenomenon of gravity: masses attract with a force proportional to Mm/r^2 . He did not explain why.

Einstein described the geometry of gravity: mass curves spacetime, and bodies follow geodesics through the curved manifold. He explained how the phenomenon arises geometrically but did not explain why mass curves spacetime.

The radiation-field model proposes the source: mass is condensed radiation; condensed radiation perturbs the ambient radiation field; the perturbation, propagating through three-dimensional space, creates an anisotropic flux that produces a directional force. The phenomenon that Newton described and the geometry that Einstein formalised are both consequences of this underlying radiative process.

The three accounts do not contradict one another. They describe the same physical reality at three levels of depth:

Level 1—Phenomenon (Newton): Masses attract with a force $F = GMm/r^2$. The inverse-square law is established empirically and mathematically, but no mechanism is proposed. The question "why do masses attract?" is explicitly set aside.

Level 2—Geometry (Einstein): Mass curves spacetime; bodies follow geodesics through the curved manifold. The phenomenon is explained geometrically, and the geometric description generates new predictions (gravitational lensing, frame-dragging, gravitational waves) that have been confirmed with extraordinary precision. But the question "why does mass curve spacetime?" receives no answer. The Einstein field equations assert the correspondence between mass-energy and curvature; they do not explain it.

Level 3—Source (SxD model): Mass is condensed radiation. Condensed radiation perturbs the ambient radiation field. The perturbation creates an anisotropic flux that pushes nearby bodies toward the shadow. In S³D, this is observed as attraction (Level 1). When the radiation field is sufficiently smooth and slowly varying, the trajectories of bodies through the perturbed field can be equivalently described as geodesics of an effective metric (Level 2). The three levels are not competing theories but nested descriptions of the same process, each complete at its own level of analysis.

Newton's inverse-square law is a consequence of spherical flux geometry in three dimensions. Einstein's metric curvature is the effective geometric encoding of the radiation-field configuration. The present model proposes the physical process from which both descriptions emerge.

V. Relationship to General Relativity: From Radiation Field to Effective Geometry

V.1 Mass-Energy as a Radiative Quantity in Einstein's Equations

The right-hand side of the Einstein field equations, $G_{uv} = 8\pi G T_{uv}$, is the stress-energy tensor T_{uv} , which encodes the energy density, momentum density, pressure, and stress at each point in spacetime [3,23]. In standard general relativity, T_{uv} is treated as a given—the matter-energy content of spacetime is specified, and the field equations determine the resulting geometry.

Within the present framework, T_{uv} acquires a physical interpretation that standard general relativity does not provide. If mass is condensed radiation and energy is radiation (Section II), then the stress-energy tensor is not a description of some ontologically fundamental "matter-energy" that happens to curve spacetime. It is the S^3D effective representation of the projected radiation field:

$$T_{uv} = T_{uv}[R]$$

where R denotes the full state of the projected radiation field. Energy density in T_{uv} corresponds to the local radiation condensation density. Momentum density corresponds to the directional flux of the radiation field. Pressure and stress correspond to the field's self-interaction at that location.

Consider the components of T_{uv} individually. The T_{00} component is the energy density—in the radiation-field model, this is the local density of radiative condensation. The T_{0i} components are the momentum density—the directional flux of the radiation field. The T_{ij} components encode pressure and shear stress—the self-interaction of the radiation field at that location. Each component of T_{uv} , which in standard general relativity is simply posited as the matter-energy content of spacetime, acquires in the present framework a specific physical interpretation as a property of the projected radiation field. The stress-energy tensor is not filled in by hand; it is determined by the state of R .

This reinterpretation does not alter any calculation within general relativity. Every prediction of the Einstein field equations remains unchanged, because the mathematical structure of T_{uv} is untouched. What changes is the ontological reading: T_{uv} is no longer an unexplained primitive but a representation of the underlying radiation field from which matter, energy, and gravity all emerge.

V.2 The Weak-Field Metric Correspondence

In the weak-field limit, general relativity reduces to the linearised approximation in which the metric is written as:

$$g_{uv} = \eta_{uv} + h_{uv}$$

where η_{uv} is the flat Minkowski metric and h_{uv} is a small perturbation. For a static, spherically symmetric mass, the dominant perturbation is $h_{00} = -2\Phi / c^2$, where Φ is the Newtonian gravitational potential [3].

In the radiation-field model, the Newtonian potential is identified with the radiation potential $U_R = -G_{\text{eff}} M / r$ (Section IV.3). The weak-field metric perturbation therefore becomes:

$$h_{00} = -2 U_R / c^2 = 2 G_{\text{eff}} M / (c^2 r)$$

This is identical to the standard linearised result. The radiation-field model therefore reproduces general relativity's weak-field predictions by construction.

V.3 The Non-Linear Regime and the Effective Metric

When the projected radiation density becomes large—in the vicinity of very massive bodies or in cosmological contexts—the linear relationship $L_R = \alpha \times M$ ceases to be adequate. The radiation field perturbation produced by M is no longer small relative to the background, and the radiation field's own structure contributes to the effective source. This is a non-linear regime in which a scalar potential Φ_R is insufficient to describe the full dynamics.

In S^3D , an observer embedded within this non-linear radiation environment requires a more general description of local spacetime relations. The standard tool for this description is the metric tensor g_{uv} . The present model proposes that the metric perturbation h_{uv} is a functional of the local radiation-field configuration:

$$h_{uv} = F_{uv}[R]$$

In the weak-field limit, this functional reduces to the scalar case $h_{00} = -2 U_R / c^2$, recovering the Newtonian regime. In the non-linear regime, the full tensorial structure of h_{uv} encodes the non-linear self-interaction of the radiation field—its coupling to its own condensations, the back-reaction of the condensed regions on the ambient field, and the resulting modifications to the local spacetime geometry experienced by S^3D observers.

This proposal does not claim to derive the Einstein field equations $G_{uv} = 8 \pi G T_{uv}$ from the radiation-field model. Such a derivation would require specifying the complete non-linear dynamics of the projected radiation field, including the field equations governing R and the explicit form of the functional F_{uv} . This lies beyond the scope of the present paper and constitutes the principal limitation of the current formulation (Section VII). What is claimed is more modest and more fundamental: the geometric curvature described by general relativity is not the cause of gravity but an effective description of the radiation-field asymmetry that produces gravitational phenomena. Einstein's equations accurately describe the geometry of the projected shadows on the S^3D screen; the SxD model identifies the projector.

V.4 Why the Effective Geometric Description Works

A natural question arises: if gravity is fundamentally a radiation-field effect, why does a purely geometric theory (general relativity) describe it so accurately?

The answer is that geometry is the natural language for describing the macroscopic effects of a pervasive field on the motion of objects embedded within it. When a radiation field is sufficiently

smooth and slowly varying, the trajectories of bodies through that field can be equivalently described as geodesics of an effective metric. This is not unique to gravity; analogous effective-metric descriptions arise in condensed matter physics (acoustic metrics in superfluids), optics (gradient-index media), and analogue gravity models [24]. In each case, the effective geometry is real in the sense that it accurately predicts trajectories, but it is not fundamental—it emerges from the underlying field dynamics.

General relativity, on this reading, is the supremely successful effective theory of gravitational geometry. Its accuracy in predicting perihelion precession, gravitational lensing, frame-dragging, and gravitational wave propagation reflects the fact that the effective-metric description captures the macroscopic behaviour of the radiation field with extraordinary fidelity. The question the present paper raises is not whether general relativity is correct—it is—but whether it is complete. The claim is that it describes the geometry without identifying the physical process that generates it.

V.5 Gravitational Waves as Radiation-Field Perturbations

The detection of gravitational waves by LIGO and Virgo [25] confirmed one of general relativity's most striking predictions: that perturbations in the spacetime metric propagate as waves at the speed of light. Any viable alternative to general relativity must account for this observation.

Within the radiation-field model, gravitational waves are not perturbations of spacetime geometry per se; they are propagating perturbations of the radiation field itself. When a massive system undergoes rapid acceleration—binary neutron stars spiralling inward, black holes merging—the radiation-field configuration changes dynamically. These changes propagate outward through the ambient radiation field at the speed of light, because the radiation field's characteristic propagation speed is c . An S³D observer equipped with an interferometer detects these propagating perturbations as oscillations in the effective metric—precisely what LIGO measures.

The formal correspondence is straightforward in the linearised regime: if $h_{uv} = F_{uv}[R]$, then a time-varying R produces a time-varying h_{uv} —a gravitational wave. The wave equation satisfied by h_{uv} in linearised general relativity (the transverse-traceless wave equation) would, on this account, be the S³D effective description of the underlying radiation-field wave equation. The detailed derivation of this correspondence—demonstrating that the radiation-field wave equation reduces to the linearised Einstein wave equation in the appropriate limit—is identified as a key objective for future work (Section VII.3).

What is important at this stage is the conceptual point: the existence of gravitational waves does not require that spacetime itself be a dynamical entity capable of oscillation. It requires only that the underlying physical medium—the radiation field—be capable of propagating perturbations. The effective-metric description translates these radiation-field perturbations into the language of spacetime geometry, producing the gravitational wave phenomenology that LIGO detects. The waves are real;

their interpretation as oscillations of spacetime is, on the present account, an effective description rather than a fundamental one.

VI. The Status of Quantum Gravity

The failure of quantum gravity programmes acquires a natural explanation within this framework. If gravity is not a fundamental field interaction but an emergent macroscopic effect of radiation-field asymmetry, then attempting to quantise the gravitational field would constitute a category error— analogous to attempting to quantise the pressure in a fluid rather than the molecules that compose it. Pressure is real, measurable, and obeys precise laws; but it is a statistical, emergent property of the underlying molecular dynamics, not a fundamental field admitting independent quantisation. Gravity, on the present account, occupies an analogous status [26].

The radiation field itself may admit a quantum description at S^4D —the quantum-field conversion interface in the SxD cascade [13]. The emergent gravitational effect observed in S^3D , however, is a macroscopic consequence of radiation-field geometry, not a quantum interaction between individual quanta. If this is correct, then the fundamental quantum degrees of freedom underlying gravity are not gravitational quanta (gravitons) but the quantum excitations of the radiation field at the S^4D level. The graviton, rather than being a missing particle awaiting detection, would not represent an independent fundamental degree of freedom but a macroscopic radiation-field effect misidentified as a fundamental quantum.

It is instructive to note the historical parallel. The programme of quantising gravity assumed, from its inception in the 1930s, that because electromagnetism—a fundamental field interaction—admits quantisation, gravity—also apparently a fundamental field interaction—must admit it too. But the analogy is only valid if gravity is, in fact, a fundamental field interaction. If it is instead an emergent macroscopic effect, the analogy fails at the root. The electromagnetic field is fundamental; its quantisation yields photons. The gravitational "field" may be effective; its quantisation yields nothing, because there is no fundamental field to quantise. The ninety-year null result of quantum gravity, on this reading, is not a failure of ingenuity but a correct consequence of a false premise.

This does not mean that gravitational phenomena are entirely classical. At the interface between the quantum regime (S^4D) and the macroscopic regime (S^3D), the radiation field's quantum structure may produce subtle corrections to the effective gravitational geometry—corrections that would manifest as departures from the predictions of classical general relativity in extreme environments (near black hole horizons, at very early cosmological epochs, or at the Planck scale). These corrections, if they exist, would constitute the empirical signature of the radiation field's quantum nature, without requiring the existence of a graviton.

VII. Conditions for Empirical Distinguishability, Limitations, and Future Directions

VII.1 Empirical Distinguishability

The force law derived in Section IV is formally identical to Newton's law, and the weak-field metric correspondence in Section V reproduces the linearised general-relativistic result. In the domain where these theories have been tested, the SxD radiation-field model therefore reproduces general relativity's weak-field predictions by construction [27].

The question is whether the model predicts any deviations from standard gravitational theory that could, in principle, be observed. Three conditions are identified.

Condition 1: Radiation-field anisotropy at cosmological scales.

If gravity arises from a background radiation field, then large-scale anisotropies in this field—arising from the geometry of the dimensional projection itself—should produce correlated deviations in gravitational dynamics at super-galactic scales. These deviations would manifest as residual correlations in galaxy distributions not accounted for by LCDM transfer functions, consistent with the non-vanishing correlation floor predicted in [13].

Condition 2: Mass-dependent coupling anomalies in the strong-field regime.

Postulates 1 and 2 assert strict proportionality between mass and radiative coupling in the weak-field limit. At extreme mass densities (neutron stars, black hole accretion environments), deviations from linearity may produce measurable departures from general-relativistic predictions—particularly in orbital precession rates or gravitational wave signatures that probe the strong-field regime. If the non-linear dynamics of the radiation field differ from those encoded in the Einstein field equations, these differences would appear precisely where the weak-field approximation breaks down.

Condition 3: Non-geometric gravitational effects.

If gravity is a radiation-field effect rather than a geometric property of spacetime, then in principle there exist configurations where the radiation-field asymmetry and the metric curvature diverge. Such configurations would produce gravitational effects not predicted by the Einstein equations alone. The identification of such configurations requires detailed modelling of the non-linear radiation-field dynamics and is proposed as a programme for future work.

These conditions do not constitute immediate falsification tests but define the empirical boundary of the model: if no deviation from standard gravity is ever observed under any of these conditions, the SxD radiation-field model remains empirically indistinguishable from general relativity, and its additional ontological commitments become a matter of philosophical preference rather than physical necessity.

VII.2 Principal Limitations

The present formulation has four principal limitations that must be acknowledged.

First, the non-linear dynamics of the projected radiation field have not been specified. The functional $F_{uv}[R]$ mapping the radiation-field configuration to the effective metric perturbation is posited but not derived. A complete theory would require radiation-field equations from which the Einstein field equations can be recovered as the macroscopic limit, analogous to the way in which the Navier-Stokes equations emerge from molecular dynamics. This derivation is the central open problem of the research programme.

Second, the coupling constants α and β (and therefore G_{eff}) are introduced as parameters of the model but are not derived from the SxD dimensional cascade. A complete account would derive these constants from the properties of the projection functional Pi introduced in [13].

Third, the correspondence between radiation-field wave dynamics and the linearised Einstein wave equation, proposed conceptually in Section V.5, has not yet been formally derived. General relativity predicts gravitational waves as propagating perturbations of the spacetime metric; the LIGO/Virgo detections have confirmed their existence [25]. The radiation-field model must account for these observations as propagating perturbations of the radiation field itself—radiation-field waves whose S^3D effective description corresponds to the metric perturbations detected by interferometers. This correspondence has not yet been formally established.

Fourth, the model has not yet addressed the behaviour of gravity at cosmological scales—specifically, the phenomena attributed to dark matter and dark energy. If gravity is a radiation-field effect, then anomalous gravitational dynamics at galactic and cosmological scales (flat rotation curves, accelerated expansion) may reflect properties of the radiation field that are not captured by the Newtonian or weak-field general-relativistic approximations. Whether the radiation-field model can account for these phenomena without invoking dark matter or dark energy is an open question with potentially transformative implications, but it lies beyond the scope of the present paper.

VII.3 Future Research Directions

The following directions identify areas where the present model motivates further investigation. They are not framed as objectives to be completed but as open questions that the radiation-field ontology makes it possible to pose.

The first direction concerns whether a macroscopic correspondence can be established between the projected radiation-field dynamics and the Einstein field equations. If $T_{uv} = T_{uv}[R]$ and $h_{uv} = F_{uv}[R]$, then the structure of the radiation field and the effective spacetime geometry are not independent. However, a complete derivation of the Einstein equations from the underlying radiation-field dynamics may not be a tractable objective within any current theoretical framework. As an analogy: biology knows the initial state of a fertilised egg and the outcome of a fully formed organism, yet cannot derive in full, from first principles, how a fertilised egg develops into a chick. The obstacle is not ignorance of the endpoint but the irreducible complexity of the generative dynamics between known boundaries. The value of this correspondence lies in establishing that the macroscopic gravitational

laws and the underlying radiative ontology are structurally connected, not in the expectation that the former can be generated from the latter in closed form. Derivability is not the measure of ontological truth; structural connection is.

The second direction concerns cosmological parameters. If the ambient radiation field is the residual projection medium of the SxD cascade, then its large-scale structure is not arbitrary but is determined by the geometry of the projection. This raises the question of whether cosmological parameters currently obtained by fitting observational data—the Hubble constant, the matter-radiation ratio, the cosmological constant—may instead reflect structural properties of the projection geometry. The present model motivates this question but does not yet possess the theoretical tools to resolve it.

The third direction concerns gravitational waves. Section V.5 proposed that gravitational waves are propagating perturbations of the radiation field whose S³D effective description corresponds to metric oscillations. The prediction that gravitational wave speed equals c is automatically satisfied, since the radiation field propagates at c . Whether the radiation-field wave equation can reproduce the polarisation structure, energy loss rate, and waveform morphology predicted by general relativity for binary inspirals and mergers remains an open question. Agreement would constrain the form of the radiation-field dynamics; disagreement would either challenge the model or point to new physics beyond general relativity.

The fourth direction, drawing on the analysis in [15], concerns the nature of invariant mass. If the electron's invariant mass represents the effective behaviour of a more fundamental massless field structure in a bound configuration—as proposed in that paper's concluding hypothesis—then the relationship between mass, binding energy, and the radiation field becomes a question that the present ontological framework makes formally addressable.

VIII. Conclusion: From Shadow to Source

Newton described what gravity does. Einstein described how gravity geometrically manifests. Neither theory, however, identifies a physical mechanism underlying gravitational attraction. This paper has proposed that gravity is an emergent radiative phenomenon: matter is a high-density condensation of a projected radiation field, and the gravitational interaction arises from the asymmetric radiation pressure produced when massive bodies shadow one another within an ambient radiation field. On this account, gravity is not a fundamental attractive interaction between masses, but a macroscopic consequence of the spatial redistribution of radiation.

The model develops this proposal from the dimensional structure of the SxD cascade and the mass-energy equivalence of special relativity. Matter is treated not as an ontologically independent substance, but as a condensed configuration of projected radiation. From this identification, the radiative source strength of a body and the coupling cross-section of a test body can be related to their projected masses

in the weak-field limit without introducing independent mass-dependent interaction assumptions. The inverse-square gravitational law then follows from the spherical dilution of radiation flux in three-dimensional space, yielding an effective force of the form $F = G_{\text{eff}} Mm/r^2$ and recovering the Newtonian potential. The resulting physical interpretation is radiation shadowing: a massive body reduces the incident ambient radiation flux in the direction it obstructs, creating a pressure imbalance that drives neighbouring bodies toward the region of reduced flux. The apparent attraction of gravity is therefore reinterpreted as the net result of an anisotropic radiative pressure field.

This reinterpretation extends beyond the Newtonian limit. Within the proposed ontology, the stress-energy tensor appearing in Einstein's field equations is interpreted as the S³D effective representation of the projected radiation field, while spacetime curvature is regarded as an effective geometric description of the underlying radiative configuration rather than its fundamental cause. In this sense, general relativity may describe the macroscopic geometry generated by a deeper radiative structure. The model therefore provides an ontological re-anchoring of gravitational physics: the familiar geometric description is retained at the macroscopic level while its physical origin is relocated to the dynamics of the projected radiation field.

The present theory nevertheless has a clearly defined limitation. The nonlinear dynamical equations governing the underlying radiation field have not yet been explicitly formulated, and consequently the full Einstein field equations have not been derived from first principles. Such a derivation may not constitute a tractable objective for any current theoretical framework, given the complexity of reconstructing a macroscopic spacetime description from a deeper generative structure. The appropriate future question is therefore not whether every macroscopic gravitational quantity can be calculated in closed form from the underlying radiation field, but whether a mathematically consistent correspondence can be established between the two levels of description. The present model should consequently be understood as a proposed structural connection between gravitational phenomenology and a deeper radiative ontology, rather than as a completed fundamental field theory.

If this interpretation is correct, the conceptual status of gravity changes fundamentally. Gravity does not require a hypothetical attractive substance or an independently existing gravitational field acting between masses. It is the macroscopic consequence of radiation pressure becoming spatially asymmetric through shadowing. What appears locally as attraction is therefore the directional result of an ambient field: an unshielded radiative flux pushes matter away from the illuminated side and toward the region of reduced flux.

Gravity is not a pull. It is a push—from the unshielded radiation behind you, toward the shadow in front of you.

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